

Simulation of GRF from Power Spectrum

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Contents

1	AIM	2
2	Fourier Space & Power Spectrum	2
3	The Discrete Case	5
3.1	Discretization	5
3.2	Notation	5
3.3	Discrete equations	6
3.4	Correlation Functions, Power Spectrum	8
4	Simulating GRF IN 2D Space	10
4.1	The size of box and discretization	10
4.2	Power Spectrum form	11
4.3	Units	12
5	Simulating GRF IN 3D Space	12
5.1	Units	13
6	Smoothing of Fields	13
6.1	Smoothing of fields in 2D	16
A	Discrete Fourier Transform & Statistical Homogeneity	17
A.1	Some Proofs	19
B	Additional Evaluations:	22
B.1	n - dimensional Gaussian and it's Fourier Transformation :	22
B.1.1	1D Gaussian :	22
B.1.2	Fourier Transform :	22
B.1.3	n - dimensional Gaussian :	22
B.2	Fourier Transform of Top Hat window function :	23
B.3	Bessel Functions Recursion :	23

List of Figures

1	Showing top hat smoothing over length scales R	14
2	Plot of Window function $\tilde{W}(k) = \exp\left[-\frac{k^2 R^2}{2}\right]$ for different values of R	15

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3	Plot of Window function $\tilde{W}(k) = \frac{3}{y^3} [\sin y - y \cos y]$ for different values of R	24
4	Plot of Window function $\tilde{W}(k) = \frac{2J_1(y)}{y}$ for different values of R	25

List of Tables

1	Symmetry and reality conditions for $\tilde{T}[\ell, m]$ in the 2D Discrete Fourier Transform (DFT) of a real-valued function, depending on the parity of N_x and N_y	11
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1 AIM

We would like to generate correlated Gaussian random field $T(\vec{\mathbf{x}})$ for a given power spectrum $P(\vec{\mathbf{k}})$. $T(\vec{\mathbf{x}})$ is the surface brightness temperature field which is a real field.

2 Fourier Space & Power Spectrum

Now we can go to Fourier space by transforming the field $T(\vec{\mathbf{x}})$ to $\tilde{T}(\vec{\mathbf{k}})$. They are related by the following relations:

$$T(\vec{\mathbf{x}}) = \int \frac{d^3\vec{\mathbf{k}}}{(2\pi)^3} \tilde{T}(\vec{\mathbf{k}}) \exp[i\vec{\mathbf{k}} \cdot \vec{\mathbf{x}}] \quad (1)$$

$$\tilde{T}(\vec{\mathbf{k}}) = \int d^3\vec{\mathbf{x}} T(\vec{\mathbf{x}}) \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{x}}] \quad (2)$$

As $T(\vec{\mathbf{x}})$ is a real field then as per Eq. (2) we have $\tilde{T}(-\vec{\mathbf{k}}) = [\tilde{T}(\vec{\mathbf{k}})]^*$.¹
Now the 2 point correlation function is given by :²

$$\xi(\vec{\mathbf{x}}_1, \vec{\mathbf{x}}_2) = \langle T(\vec{\mathbf{x}}_1) T(\vec{\mathbf{x}}_2) \rangle \quad (3)$$

$$\begin{aligned} &= \langle T(\vec{\mathbf{x}}_1) [T(\vec{\mathbf{x}}_2)]^* \rangle \\ &= \int \int \frac{d^3\vec{\mathbf{k}}_1}{(2\pi)^3} \frac{d^3\vec{\mathbf{k}}_2}{(2\pi)^3} \langle \tilde{T}(\vec{\mathbf{k}}_1) [\tilde{T}(\vec{\mathbf{k}}_2)]^* \rangle \exp[i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1 - i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \end{aligned} \quad (4)$$

Now, in our context we assume that our field is statistically homogeneous. Meaning no preferred origin. So the correlation function should be a function of only the separation vector, i.e. $\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2$. Implying $\xi(\vec{\mathbf{x}}_1, \vec{\mathbf{x}}_2) = \xi(\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2)$.

Now What Happens in Fourier Space ?

$$\langle \tilde{T}(\vec{\mathbf{k}}_1) [\tilde{T}(\vec{\mathbf{k}}_2)]^* \rangle = \int \int d^3\vec{\mathbf{x}}_1 d^3\vec{\mathbf{x}}_2 \langle T(\vec{\mathbf{x}}_1) T(\vec{\mathbf{x}}_2) \rangle \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1 + i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \quad (5)$$

$$\begin{aligned} &= \int \int d^3\vec{\mathbf{x}}_1 d^3\vec{\mathbf{x}}_2 \xi(\vec{\mathbf{x}}_1, \vec{\mathbf{x}}_2) \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1 + i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \\ &= \int \int d^3\vec{\mathbf{x}}_1 d^3\vec{\mathbf{x}}_2 \xi(\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2) \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1 + i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \\ &= \int d^3\vec{\mathbf{x}}_2 \exp[i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \int d^3\vec{\mathbf{x}}_1 \xi(\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2) \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1] \end{aligned} \quad (6)$$

¹ Where ‘*’ means complex conjugation

² Where $\langle \dots \rangle$ represents ensemble average.

Make the substitution $\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2 = \vec{\mathbf{r}}$. For fixed value of $\vec{\mathbf{x}}_2$, we have $d^3\vec{\mathbf{x}}_1 = d^3\vec{\mathbf{r}}$.

So the last integral becomes :

$$\begin{aligned} \langle \tilde{T}(\vec{\mathbf{k}}_1) [\tilde{T}(\vec{\mathbf{k}}_2)]^* \rangle &= \int d^3\vec{\mathbf{x}}_2 \exp[i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \int d^3\vec{\mathbf{r}} \xi(\vec{\mathbf{r}}) \exp[-i\vec{\mathbf{k}}_1 \cdot (\vec{\mathbf{x}}_2 + \vec{\mathbf{r}})] \\ &= \int d^3\vec{\mathbf{r}} \xi(\vec{\mathbf{r}}) \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{r}}] \int d^3\vec{\mathbf{x}}_2 \exp[i(\vec{\mathbf{k}}_2 - \vec{\mathbf{k}}_1) \cdot \vec{\mathbf{x}}_2] \\ &= \int d^3\vec{\mathbf{r}} \xi(\vec{\mathbf{r}}) \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{r}}] \times (2\pi)^3 \delta_D^3(\vec{\mathbf{k}}_1 - \vec{\mathbf{k}}_2) \quad ^1 \\ &= (2\pi)^3 \delta_D^3(\vec{\mathbf{k}}_1 - \vec{\mathbf{k}}_2) P(\vec{\mathbf{k}}_1) \end{aligned} \quad (7)$$

We finally have :

$$\langle \tilde{T}(\vec{\mathbf{k}}_1) [\tilde{T}(\vec{\mathbf{k}}_2)]^* \rangle = (2\pi)^3 \delta_D^3(\vec{\mathbf{k}}_1 - \vec{\mathbf{k}}_2) P(\vec{\mathbf{k}}_1) \quad ^2 \quad (8)$$

And also :

$$P(\vec{\mathbf{k}}) = \int d^3\vec{\mathbf{r}} \xi(\vec{\mathbf{r}}) \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \quad (9)$$

Now using Eq. (4) and Eq. (8) we have:

$$\begin{aligned} \xi(\vec{\mathbf{r}}) &= \int \int \frac{d^3\vec{\mathbf{k}}_1}{(2\pi)^3} \frac{d^3\vec{\mathbf{k}}_2}{(2\pi)^3} \left\{ (2\pi)^3 \delta_D^3(\vec{\mathbf{k}}_1 - \vec{\mathbf{k}}_2) P(\vec{\mathbf{k}}_1) \right\} \exp[i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1 - i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \\ &= \int \frac{d^3\vec{\mathbf{k}}_1}{(2\pi)^3} P(\vec{\mathbf{k}}_1) \exp[i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1] \int d^3\vec{\mathbf{k}}_2 \delta_D^3(\vec{\mathbf{k}}_1 - \vec{\mathbf{k}}_2) \exp[-i\vec{\mathbf{k}}_2 \cdot \vec{\mathbf{x}}_2] \\ &= \int \frac{d^3\vec{\mathbf{k}}_1}{(2\pi)^3} P(\vec{\mathbf{k}}_1) \exp[i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_1] \exp[-i\vec{\mathbf{k}}_1 \cdot \vec{\mathbf{x}}_2] \\ &= \int \frac{d^3\vec{\mathbf{k}}_1}{(2\pi)^3} P(\vec{\mathbf{k}}_1) \exp[i\vec{\mathbf{k}}_1 \cdot (\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2)] \end{aligned}$$

So we finally have :

$$\xi(\vec{\mathbf{r}}) = \int \frac{d^3\vec{\mathbf{k}}}{(2\pi)^3} P(\vec{\mathbf{k}}) \exp[i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \quad (10)$$

So by Eq. (9) and Eq. (10) we can see that $P(\vec{\mathbf{k}})$ and $\xi(\vec{\mathbf{r}})$ are the Fourier transform pairs. This is the result of statistically homogenous fields and this theorem is known as **Wiener-Khinchin-Einstein Theorem**.

Now $P(\vec{\mathbf{k}})$ is known as power spectrum. This measures the amplitude square of each Fourier mode $\vec{\mathbf{k}}$. Now if we impose the statistical isotropy, meaning no preferred direction, as a result the correlation function is only a function of absolute distance. Implying :

$$\xi(\vec{\mathbf{x}}_1, \vec{\mathbf{x}}_2) = \xi(\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2) = \xi(|\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2|) \quad (11)$$

¹ Where δ_D^3 represents three dimensional Dirac delta function. Also we have used :

$$\delta_D^3(\vec{\mathbf{k}}) = \int \frac{d^3\vec{\mathbf{x}}}{(2\pi)^3} \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{x}}]$$

This comes from Eq. (1) and Eq. (2). If we take the Fourier transform of $\delta_D^3(\vec{\mathbf{k}})$ we get the value $1/(2\pi)^3$ (by Eq. (1)). Now the inverse Fourier transform of $\delta_D^3(\vec{\mathbf{k}})$ would be given by putting $T(\vec{\mathbf{x}}) = 1/(2\pi)^3$ in Eq. (2). Putting that we get the relation stated earlier.

² This tells us that due to presence of Dirac delta function, in Fourier space the fields get decoupled. (becomes uncorrelated)

So by virtue of Eq. (9) we have $P(\vec{\mathbf{k}}) = P(|\vec{\mathbf{k}}|) = P(k)$. The isotropic power spectrum.¹ This follows from here :

$$\begin{aligned}
 P(\vec{\mathbf{k}}) &= \int d^3\vec{\mathbf{r}} \xi(\vec{\mathbf{r}}) \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \\
 &= \int d^3\vec{\mathbf{r}} \xi(r) \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \quad (\text{Using Isotropy}) \\
 &= \iiint r^2 dr \sin \theta d\theta d\phi \xi(r) \exp[-ikr \cos \theta] \\
 &= (2\pi) \int_0^\infty \xi(r) r^2 dr \int_0^\pi \exp[-ikr \cos \theta] \sin \theta d\theta \\
 &= 4\pi \int_0^\infty r^2 dr \xi(r) \frac{\sin(kr)}{kr} \quad ^3 \\
 &= P(k)
 \end{aligned}$$

Summary:

$$\begin{aligned}
 T(\vec{\mathbf{x}}) &= \int \frac{d^3\vec{\mathbf{k}}}{(2\pi)^3} \tilde{T}(\vec{\mathbf{k}}) \exp[i\vec{\mathbf{k}} \cdot \vec{\mathbf{x}}] \\
 \tilde{T}(\vec{\mathbf{k}}) &= \int d^3\vec{\mathbf{x}} T(\vec{\mathbf{x}}) \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{x}}] \\
 \langle \tilde{T}(\vec{\mathbf{k}}_1) [\tilde{T}(\vec{\mathbf{k}}_2)]^* \rangle &= (2\pi)^3 \delta_D^3(\vec{\mathbf{k}}_1 - \vec{\mathbf{k}}_2) P(\vec{\mathbf{k}}_1) \\
 P(\vec{\mathbf{k}}) &= \int d^3\vec{\mathbf{r}} \xi(\vec{\mathbf{r}}) \exp[-i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \\
 \xi(\vec{\mathbf{r}}) &= \int \frac{d^3\vec{\mathbf{k}}}{(2\pi)^3} P(\vec{\mathbf{k}}) \exp[i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \\
 \xi(\vec{\mathbf{r}}) &= \langle T(\vec{\mathbf{x}} + \vec{\mathbf{r}}) T(\vec{\mathbf{x}}) \rangle
 \end{aligned}$$

Notes :

¹ Also true for the other way round. If power spectrum $P(\vec{\mathbf{k}})$ is isotropic i.e. $P(k)$ then by virtue of Eq. (10) we have $\xi(\vec{\mathbf{r}}) = \xi(r)$.²

² The same way we have

$$\xi(r) = \frac{1}{2\pi^2} \int_0^\infty k^2 dk P(k) \frac{\sin(kr)}{kr}$$

³ Where we have used

$$\begin{aligned}
 &\int_0^\pi \exp[-ikr \cos \theta] \sin \theta d\theta \\
 &= \int_{-1}^1 \exp[-ikr \cos \theta] d[\cos(\theta)] \\
 &= \frac{\exp[-ikr \cos \theta]}{-ikr} \Big|_{-1}^{+1} \\
 &= \frac{\exp[-ikr] - \exp[ikr]}{-ikr} \\
 &= \frac{-2i \sin kr}{-ikr} \\
 &= \frac{2 \sin(kr)}{kr}
 \end{aligned}$$

If $P(\vec{\mathbf{k}}) = 1$ then from this, generated field $T(\vec{\mathbf{x}})$ is called the **white noise**. As you can see for $P(\vec{\mathbf{k}}) = 1$, $\xi(\vec{\mathbf{r}}) = \int \frac{d^3\vec{\mathbf{k}}}{(2\pi)^3} \exp[i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] = \delta^3(\vec{\mathbf{r}})$. So it gives delta covariance in real space. So fields are uncorrelated in real space. That's why it is called the white noise.

3 The Discrete Case

When we simulate the fields we don't have the continuous data points, instead we have discrete set of points. We have a finite spatial extent. Due to this, in Fourier space the modes $\vec{\mathbf{k}}$ are not longer continuous, rather than they are discrete.¹

3.1 Discretization

We take a finite box of volume V , extending from 0 to L_x in x direction, 0 to L_y in y direction and 0 to L_z in z direction. We take N_x, N_y, N_z no of grid points along the x, y, z direction respectively. So the spacing is $\Delta x = \frac{L_x}{N_x}, \Delta y = \frac{L_y}{N_y}, \Delta z = \frac{L_z}{N_z}$ Where $V = L_x L_y L_z$. Due to this discretization, the wave-vector $\vec{\mathbf{k}}$ gets discretized.

The components of $\vec{\mathbf{k}}$ are (k_x, k_y, k_z) . Now their components are given by :

$$k_i = \frac{2\pi m_i}{L_i} \quad \text{for } i = x, y, z$$

Where $m_i = [0, 1, \dots, N_i - 1]$, gives $\Delta k_x = \frac{2\pi}{L_x}, \Delta k_y = \frac{2\pi}{L_y}, \Delta k_z = \frac{2\pi}{L_z}$. Now due to sampling the modes can't have frequencies higher than Nyquist frequency. So the Nyquist frequency is given by $\frac{\pi N_i}{L_i}$. Then we have :

$$k_x = \begin{cases} \frac{2\pi}{L_x} m_x & \text{for } 0 \leq m_x \leq \frac{N}{2} \\ \frac{2\pi}{L_x} (m_x - N) & \text{for } m_x > \frac{N}{2} \end{cases} \quad (12)$$

Same goes for k_y and k_z and m_x, m_y, m_z are integers.

For example :

$$k_x = \begin{cases} \frac{2\pi}{L_x} [0, 1, \dots, \frac{N_x-1}{2}, \frac{N_x}{2}, -\frac{N_x-1}{2}, -\frac{N_x-2}{2}, \dots, -2, -1] & \text{for } N_x \text{ even.} \\ \frac{2\pi}{L_x} [0, 1, \dots, \frac{N_x-1}{2}, -\frac{N_x-1}{2}, -\frac{N_x-2}{2}, \dots, -2, -1] & \text{for } N_x \text{ odd.} \end{cases} \quad (13)$$

3.2 Notation

We can use a compact notation, where we write a position vector $\vec{\mathbf{x}}_{\mathbf{m}} = (x_p, y_q, z_r)$, $\mathbf{m} \rightarrow p, q, r$ represent the real space grid point indices. And also in same way $\vec{\mathbf{k}}_{\mathbf{n}} = (\frac{2\pi n_x}{L_x}, \frac{2\pi n_y}{L_y}, \frac{2\pi n_z}{L_z})$. $\mathbf{n} \rightarrow \ell, m, n$ represent Fourier space grid points.

Where :

$$\begin{aligned} p, \ell &= 0, 1, \dots, N_x - 1 \\ q, m &= 0, 1, \dots, N_y - 1 \\ r, n &= 0, 1, \dots, N_z - 1 \end{aligned}$$

¹ For more details see Appendix appendix A.

And the integers n_x, n_y, n_z are given by :

$$\begin{aligned} n_x &= \begin{cases} 0, 1, \dots, \frac{N_x-1}{2}, \frac{N_x}{2}, -\frac{N_x-1}{2}, -\frac{N_x-2}{2}, \dots, -2, -1 & \text{if } N_x \text{ is even.} \\ 0, 1, \dots, \frac{N_x-1}{2}, -\frac{N_x-1}{2}, -\frac{N_x-2}{2}, \dots, -2, -1 & \text{if } N_x \text{ is odd.} \end{cases} \\ n_y &= \begin{cases} 0, 1, \dots, \frac{N_y-1}{2}, \frac{N_y}{2}, -\frac{N_y-1}{2}, -\frac{N_y-2}{2}, \dots, -2, -1 & \text{if } N_y \text{ is even.} \\ 0, 1, \dots, \frac{N_y-1}{2}, -\frac{N_y-1}{2}, -\frac{N_y-2}{2}, \dots, -2, -1 & \text{if } N_y \text{ is odd.} \end{cases} \\ n_z &= \begin{cases} 0, 1, \dots, \frac{N_z-1}{2}, \frac{N_z}{2}, -\frac{N_z-1}{2}, -\frac{N_z-2}{2}, \dots, -2, -1 & \text{if } N_z \text{ is even.} \\ 0, 1, \dots, \frac{N_z-1}{2}, -\frac{N_z-1}{2}, -\frac{N_z-2}{2}, \dots, -2, -1 & \text{if } N_z \text{ is odd.} \end{cases} \end{aligned}$$

In short :

$$\begin{aligned} n_x &= \begin{cases} \ell & \text{for } 0 \leq \ell \leq \frac{N_x}{2} \\ \ell - N_x & \text{for } \ell > \frac{N_x}{2} \end{cases} \\ n_y &= \begin{cases} m & \text{for } 0 \leq m \leq \frac{N_y}{2} \\ m - N_y & \text{for } m > \frac{N_y}{2} \end{cases} \\ n_z &= \begin{cases} n & \text{for } 0 \leq n \leq \frac{N_z}{2} \\ n - N_z & \text{for } n > \frac{N_z}{2} \end{cases} \end{aligned}$$

3.3 Discrete equations

Now the discrete version of Eq. (1) is given by :

$$T(\vec{\mathbf{x}}_{\mathbf{m}}) = \frac{1}{V} \sum_{\mathbf{n}} \tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) \exp[i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}}] \quad (14)$$

and also for Eq. (2) is :

$$\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) = \Delta V \sum_{\mathbf{m}} T(\vec{\mathbf{x}}_{\mathbf{m}}) \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}}] \quad (15)$$

Where we have replaced :

$$\begin{aligned} \int d^3\vec{\mathbf{x}} &\rightarrow \sum_{\mathbf{m}} \Delta V \\ \int \frac{d^3\vec{\mathbf{k}}}{(2\pi)^3} &\rightarrow \sum_{\mathbf{n}} \frac{1}{V} \end{aligned} \quad \text{¹}$$

¹ The integrals have become summation and

$$\begin{aligned} d^3\vec{\mathbf{x}} &= \Delta x \Delta y \Delta z = \frac{V}{N_x N_y N_z} = \Delta V \\ d^3\vec{\mathbf{k}} &= \Delta k_x \Delta k_y \Delta k_z = \left(\frac{2\pi}{L_x}\right) \left(\frac{2\pi}{L_y}\right) \left(\frac{2\pi}{L_z}\right) = \frac{(2\pi)^3}{V} \end{aligned}$$

To be clear $\sum_{\mathbf{m}}$ indicates sum over all p, q, r , i.e. $\sum_{\mathbf{m}} \equiv \sum_{p=0}^{N_x-1} \sum_{q=0}^{N_y-1} \sum_{r=0}^{N_z-1}$ and same goes for $\sum_{\mathbf{n}}$ summing over ℓ, m, n (all possible values)

$\mathbf{n}$ refers to grid point in the Fourier space have wave-vector $\vec{\mathbf{k}}_{\mathbf{n}} = \left(\frac{2\pi n_x}{L_x}, \frac{2\pi n_y}{L_y}, \frac{2\pi n_z}{L_z}\right)$.

Most important thing here, discrete Fourier transformation assumes periodicity. $T(\vec{x}_m)$, $\tilde{T}(\vec{k}_n)$ are assumed to be periodic outside our defined grid.¹ Also in this discretized version we have the property that

$$\left[\tilde{T}(\vec{k}_n)\right]^* = \tilde{T}(\vec{k}_{N-n}) = \tilde{T}(\vec{k}_{-n})$$
²

¹ So we assume that $T(\vec{x}_m) = T(\vec{x}_{m+N})$, $\tilde{T}(\vec{k}_n) = \tilde{T}(\vec{k}_{n+N})$ for $\mathbf{N} = (N_x, N_y, N_z)$.

² We refer to $\vec{k}_{-n}$ be same as $\vec{k}_{N-n}$ due to periodicity.

Also by Eq. (15) :

$$\begin{aligned} \tilde{T}(\vec{k}_{N-n}) &= \Delta V \sum_{\mathbf{m}} T(\vec{x}_m) \exp \left[-i \vec{k}_{N-n} \cdot \vec{x}_m \right] \\ &= \Delta V \sum_{\mathbf{m}} T(\vec{x}_m) \exp \left[i \vec{k}_n \cdot \vec{x}_m \right] \text{³} = \left\{ \Delta V \sum_{\mathbf{m}} T(\vec{x}_m) \exp \left[-i \vec{k}_n \cdot \vec{x}_m \right] \right\}^* = \left[\tilde{T}(\vec{k}_n) \right]^* \end{aligned}$$

³ See Appendix appendix A.1 for proof.

3.4 Correlation Functions, Power Spectrum

As we have done in the continuous case, we will do the same thing here for the discrete case. 2 point correlation function is given by :

$$\begin{aligned}\xi(\vec{\mathbf{x}}_{\mathbf{m}}, \vec{\mathbf{x}}_{\mathbf{m}'}) &= \langle T(\vec{\mathbf{x}}_{\mathbf{m}}) T(\vec{\mathbf{x}}_{\mathbf{m}'}) \rangle \\ &= \langle T(\vec{\mathbf{x}}_{\mathbf{m}}) [T(\vec{\mathbf{x}}_{\mathbf{m}'})]^* \rangle \\ &= \frac{1}{V^2} \sum_{\mathbf{n}} \sum_{\mathbf{n}'} \langle \tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) [\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}'})]^* \rangle \exp[i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}} - i\vec{\mathbf{k}}_{\mathbf{n}'} \cdot \vec{\mathbf{x}}_{\mathbf{m}'}] \end{aligned} \quad (16)$$

Using statistical homogeneity, we have :

$$\xi(\vec{\mathbf{x}}_{\mathbf{m}}, \vec{\mathbf{x}}_{\mathbf{m}'}) = \xi(\vec{\mathbf{x}}_{\mathbf{m}} - \vec{\mathbf{x}}_{\mathbf{m}'})$$

Now What Happens in Fourier Space ?

$$\begin{aligned}\langle \tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) [\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}'})]^* \rangle &= (\Delta V)^2 \sum_{\mathbf{m}} \sum_{\mathbf{m}'} \langle T(\vec{\mathbf{x}}_{\mathbf{m}}) [T(\vec{\mathbf{x}}_{\mathbf{m}'})]^* \rangle \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}} + i\vec{\mathbf{k}}_{\mathbf{n}'} \cdot \vec{\mathbf{x}}_{\mathbf{m}'}] \\ &= (\Delta V)^2 \sum_{\mathbf{m}} \sum_{\mathbf{m}'} \xi(\vec{\mathbf{x}}_{\mathbf{m}} - \vec{\mathbf{x}}_{\mathbf{m}'}) \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}} + i\vec{\mathbf{k}}_{\mathbf{n}'} \cdot \vec{\mathbf{x}}_{\mathbf{m}'}] \\ &= (\Delta V)^2 \sum_{\mathbf{m}'} \exp[i\vec{\mathbf{k}}_{\mathbf{n}'} \cdot \vec{\mathbf{x}}_{\mathbf{m}'}] \sum_{\mathbf{m}} \xi(\vec{\mathbf{x}}_{\mathbf{m}} - \vec{\mathbf{x}}_{\mathbf{m}'}) \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}}] \end{aligned} \quad (17)$$

Make the substitution $\vec{\mathbf{x}}_{\mathbf{m}} - \vec{\mathbf{x}}_{\mathbf{m}'} = \vec{\mathbf{x}}_{\mathbf{m}''}$.¹ So the term becomes :

$$\begin{aligned}\langle \tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) [\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}'})]^* \rangle &= (\Delta V)^2 \sum_{\mathbf{m}'} \exp[i\vec{\mathbf{k}}_{\mathbf{n}'} \cdot \vec{\mathbf{x}}_{\mathbf{m}'}] \sum_{\mathbf{m}''} \xi(\vec{\mathbf{x}}_{\mathbf{m}''}) \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot (\vec{\mathbf{x}}_{\mathbf{m}'} + \vec{\mathbf{x}}_{\mathbf{m}''})] \\ &= (\Delta V)^2 \sum_{\mathbf{m}''} \xi(\vec{\mathbf{x}}_{\mathbf{m}''}) \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}''}] \sum_{\mathbf{m}'} \exp[i(\vec{\mathbf{k}}_{\mathbf{n}'} - \vec{\mathbf{k}}_{\mathbf{n}}) \cdot \vec{\mathbf{x}}_{\mathbf{m}'}] \\ &= (\Delta V)^2 \left\{ \sum_{\mathbf{m}''} \xi(\vec{\mathbf{x}}_{\mathbf{m}''}) \exp[-i\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}''}] \right\} \times \left\{ N_x N_y N_z \delta_{\vec{\mathbf{k}}_{\mathbf{n}}, \vec{\mathbf{k}}_{\mathbf{n}'}} \right\}^2 \end{aligned} \quad (18)$$

¹ Does this substitution right? Are we losing any term or just wrong ? See Appendix appendix A.

² Where $\delta_{\vec{\mathbf{k}}_{\mathbf{n}}, \vec{\mathbf{k}}_{\mathbf{n}'}}$ is the Kronecker delta. See Appendix appendix A.1 for evaluation.

Kronecker delta is defined as :

$$\delta_{\vec{\mathbf{k}}_{\mathbf{n}}, \vec{\mathbf{k}}_{\mathbf{n}'}} = \begin{cases} 1 & \text{if } \vec{\mathbf{k}}_{\mathbf{n}} = \vec{\mathbf{k}}_{\mathbf{n}'} \\ 0 & \text{if } \vec{\mathbf{k}}_{\mathbf{n}} \neq \vec{\mathbf{k}}_{\mathbf{n}'} \end{cases}$$

Eq. (18) can be slightly arranged to write :

$$\begin{aligned}
 \langle \tilde{T}(\vec{k}_n) [\tilde{T}(\vec{k}_{n'})]^* \rangle &= \Delta V \left\{ \Delta V \sum_{\vec{m}''} \xi(\vec{x}_{m''}) \exp[-i \vec{k}_n \cdot \vec{x}_{m''}] \right\} \times \left\{ N_x N_y N_z \delta_{\vec{k}_n, \vec{k}_{n'}} \right\} \\
 &= V \delta_{\vec{k}_n, \vec{k}_{n'}} \left\{ \Delta V \sum_{\vec{m}''} \xi(\vec{x}_{m''}) \exp[-i \vec{k}_n \cdot \vec{x}_{m''}] \right\} \\
 &= V \delta_{\vec{k}_n, \vec{k}_{n'}} P(\vec{k}_n)
 \end{aligned}$$

So we have some formulae :

$$\langle \tilde{T}(\vec{k}_n) [\tilde{T}(\vec{k}_{n'})]^* \rangle = V \delta_{\vec{k}_n, \vec{k}_{n'}} P(\vec{k}_n) \quad (19)$$

$$P(\vec{k}_n) = \frac{1}{V} \langle |\tilde{T}(\vec{k}_n)|^2 \rangle \quad (20)$$

$$P(\vec{k}_n) = \Delta V \sum_{\vec{m}''} \xi(\vec{x}_{m''}) \exp[-i \vec{k}_n \cdot \vec{x}_{m''}] \quad (21)$$

Now using Eq. (19) in Eq. (16) we have :

$$\begin{aligned}
 \xi(\vec{x}_{m''}) &= \frac{1}{V^2} \sum_n \sum_{n'} \langle \tilde{T}(\vec{k}_n) [\tilde{T}(\vec{k}_{n'})]^* \rangle \exp[i \vec{k}_n \cdot \vec{x}_m - i \vec{k}_{n'} \cdot \vec{x}_{m'}] \\
 &= \frac{1}{V^2} \sum_n \sum_{n'} V \delta_{\vec{k}_n, \vec{k}_{n'}} P(\vec{k}_n) \exp[i \vec{k}_n \cdot \vec{x}_m - i \vec{k}_{n'} \cdot \vec{x}_{m'}] \\
 &= \frac{1}{V} \sum_n P(\vec{k}_n) \exp[i \vec{k}_n \cdot \vec{x}_m] \sum_{n'} \delta_{\vec{k}_n, \vec{k}_{n'}} \exp[-i \vec{k}_{n'} \cdot \vec{x}_{m'}] \\
 &= \frac{1}{V} \sum_n P(\vec{k}_n) \exp[i \vec{k}_n \cdot \vec{x}_m] \exp[-i \vec{k}_n \cdot \vec{x}_{m'}] \\
 &= \frac{1}{V} \sum_n P(\vec{k}_n) \exp[i \vec{k}_n \cdot (\vec{x}_m - \vec{x}_{m'})]
 \end{aligned}$$

$$\xi(\vec{x}_{m''}) = \frac{1}{V} \sum_n P(\vec{k}_n) \exp[i \vec{k}_n \cdot \vec{x}_{m''}] \quad (22)$$

By Eq. (21) and Eq. (22) we can see that $\xi(\vec{x}_m)$ and $P(\vec{k}_n)$ are the Fourier conjugate pairs. (as expected).

Summary:

$$\begin{aligned}
 T(\vec{x}_m) &= \frac{1}{V} \sum_n \tilde{T}(\vec{k}_n) \exp[i \vec{k}_n \cdot \vec{x}_m] \\
 \tilde{T}(\vec{k}_n) &= \Delta V \sum_m T(\vec{x}_m) \exp[-i \vec{k}_n \cdot \vec{x}_m] \\
 \langle \tilde{T}(\vec{k}_n) [\tilde{T}(\vec{k}_{n'})]^* \rangle &= V \delta_{\vec{k}_n, \vec{k}_{n'}} P(\vec{k}_n) \\
 P(\vec{k}_n) &= \frac{1}{V} \langle |\tilde{T}(\vec{k}_n)|^2 \rangle \\
 P(\vec{k}_n) &= \Delta V \sum_{\vec{m}''} \xi(\vec{x}_{m''}) \exp[-i \vec{k}_n \cdot \vec{x}_{m''}] \\
 \xi(\vec{x}_{m''}) &= \frac{1}{V} \sum_n P(\vec{k}_n) \exp[i \vec{k}_n \cdot \vec{x}_{m''}]
 \end{aligned}$$

4 Simulating GRF IN 2D Space

Now here comes the most important part. A correlated GRF can be simulated by taking Fourier transform of uncorrelated GRF in Fourier space.

4.1 The size of box and discretization

In 2D space we take a rectangular box having N_x, N_y no of grid points along x and y direction respectively with spacing $\Delta x = \frac{L_x}{N_x}, \Delta y = \frac{L_y}{N_y}$. In Fourier space we have spacing $\Delta k_x = \frac{2\pi}{L_x}, \Delta k_y = \frac{2\pi}{L_y}$, with area $A = L_x L_y$.

Now we denote $\vec{\mathbf{x}}_{\mathbf{m}} = (p\Delta x, q\Delta y)$ with $\mathbf{m}$ representing the indices p, q . The same way $\vec{\mathbf{k}}_{\mathbf{n}} = (n_x\Delta k_x, n_y\Delta k_y)$ with $\mathbf{n}$ representing indices ℓ, m where :

$$n_x = \begin{cases} \ell & \text{for } 0 \leq \ell \leq \frac{N_x}{2} \\ \ell - N_x & \text{for } \ell > \frac{N_x}{2} \end{cases}$$

$$n_y = \begin{cases} m & \text{for } 0 \leq m \leq \frac{N_y}{2} \\ m - N_y & \text{for } m > \frac{N_y}{2} \end{cases}$$

Also for representing the field we have :

$$T(\vec{\mathbf{x}}_{\mathbf{m}}) = T[p, q]$$

$$\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) = \tilde{T}[\ell, m]$$

So the discrete versions of Eq. (1) and Eq. (2) in 2D are :

$$T(\vec{\mathbf{x}}_{\mathbf{m}}) = \frac{1}{A} \sum_{\mathbf{n}} \tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) \exp[i \vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}}]$$

$$\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) = \Delta A \sum_{\mathbf{m}} T(\vec{\mathbf{x}}_{\mathbf{m}}) \exp[-i \vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}}]$$

By our convention this becomes :

$$T[p, q] = \frac{1}{A} \sum_{\ell, m} \tilde{T}[\ell, m] \exp\left[2\pi i \left(\frac{\ell p}{N_x} + \frac{mq}{N_y}\right)\right]$$

$$\tilde{T}[\ell, m] = \Delta A \sum_{p, q} T[p, q] \exp\left[-2\pi i \left(\frac{\ell p}{N_x} + \frac{mq}{N_y}\right)\right]^1$$

¹ Where, we have written

$$\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}} = 2\pi i \left(\frac{\ell p}{N_x} + \frac{mq}{N_y}\right)$$

With $\Delta A = \frac{A}{N_x N_y}$

Now Eq. (20) gives

$$\langle |\tilde{T}[\ell, m]|^2 \rangle = A P(\vec{k}_n) \quad (23)$$

If we assume isotropy then the last equation becomes :

$$\langle |\tilde{T}[\ell, m]|^2 \rangle = A P(k_n) \quad (24)$$

So this equation tells us $\tilde{T}[\ell, m]$ has variance $A P(k_n)$ and as it is a complex field we decompose it in this way :

$$\tilde{T}[\ell, m] = \sqrt{\frac{A P(k_n)}{2}} [a(\ell, m) + ib(\ell, m)] \quad (25)$$

Where we take $a(\ell, m), b(\ell, m)$ from a normal distribution $\sim \mathcal{N}(0, 1)$ with zero mean and unit variance. But as $T[p, q]$ is a real field we get the symmetry property:

$$\tilde{T}[N_x - \ell, N_y - m] = [\tilde{T}[\ell, m]]^* \quad (26)$$

Where ℓ, m runs from $0, N_x - 1$ and $0, N_y - 1$ respectively. But this gives a condition :

$$\tilde{T}[N_x, N_y] = \tilde{T}[0, 0] = [\tilde{T}[0, 0]]^* \quad ^1$$

So this states $\tilde{T}[0, 0]$ should be real.

Now 4 cases can arise. We summarize in table Table. 1.

N_x, N_y both odd	N_x odd , N_y even	N_x even , N_y odd	N_x, N_y both even
No match other than $\ell = 0, m = 0$	Match on $\ell = 0, m = N_y/2$	Match on $\ell = N_x/2, m = 0$	Match on: $\begin{cases} \ell = 0, m = N_y/2 \\ \ell = N_x/2, m = 0 \\ \ell = N_x/2, m = N_y/2 \end{cases}$
–	$\tilde{T}[0, N_y/2]$ is real	$\tilde{T}[N_x/2, 0]$ is real	$\begin{cases} \tilde{T}[0, N_y/2] \text{ is real} \\ \tilde{T}[N_x/2, 0] \text{ is real} \\ \tilde{T}[N_x/2, N_y/2] \text{ is real} \end{cases}$

Table 1: Symmetry and reality conditions for $\tilde{T}[\ell, m]$ in the 2D Discrete Fourier Transform (DFT) of a real-valued function, depending on the parity of N_x and N_y .

If we do have the reality condition then we generate $\tilde{T}[\ell, m]$ by:

$$\tilde{T}[\ell, m] = \sqrt{A P(k_n)} [a(\ell, m)]$$

Where we take $a(\ell, m)$ from a normal distribution $\sim \mathcal{N}(0, 1)$ with zero mean and unit variance.

4.2 Power Spectrum form

Now we take the power spectrum to be isotopic and take the form as of $P(k)$ as power law:

$$P(k) = A_s \left(\frac{k}{k_0} \right)^s \quad (27)$$

Where A_s defines the amplitude and s is the power index. We have set, for $k = 0$ $P(k) = 0$ meaning zero mean GRF field in real space.

¹ Putting $\ell, m = 0, 0$ in Eq. (26) and using periodicity.

4.3 Units

$T(\vec{\mathbf{x}}_{\mathbf{m}})$ has unit of temperature K (kelvin) then $\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}})$ has the unit of $K \text{ Mpc}^2$. Where we measure the distances in Mpc unit. Now power spectrum has the unit of $K^2 \text{ Mpc}^2$. So the amplitude A_s has the unit of $K^2 \text{ Mpc}^2$. and we have taken $k_0 = 1 \text{ Mpc}^{-1}$.

5 Simulating GRF IN 3D Space

The same way we can do the simulations in 3D space. I will just outline the formulae.

In 3D space we take a box having N_x, N_y, N_z no of grid points along x, y and z direction respectively with spacing $\Delta x = \frac{L_x}{N_x}, \Delta y = \frac{L_y}{N_y}, \Delta z = \frac{L_z}{N_z}$. In Fourier space we have spacing $\Delta k_x = \frac{2\pi}{L_x}, \Delta k_y = \frac{2\pi}{L_y}, \Delta k_z = \frac{2\pi}{L_z}$, with volume $V = L_x L_y L_z$.

Now we denote $\vec{\mathbf{x}}_{\mathbf{m}} = (p\Delta x, q\Delta y, r\Delta z)$ with $\mathbf{m}$ representing the indices p, q, r . The same way $\vec{\mathbf{k}}_{\mathbf{n}} = (n_x\Delta k_x, n_y\Delta k_y, n_z\Delta k_z)$ with $\mathbf{n}$ representing indices ℓ, m, n . Also for representing the field we have :

$$T(\vec{\mathbf{x}}_{\mathbf{m}}) = T[p, q, r]$$

$$\tilde{T}(\vec{\mathbf{k}}_{\mathbf{n}}) = \tilde{T}[\ell, m, n]$$

By our convention this becomes :

$$T[p, q, r] = \frac{1}{V} \sum_{\ell, m, n} \tilde{T}[\ell, m, n] \exp \left[2\pi i \left(\frac{\ell p}{N_x} + \frac{mq}{N_y} + \frac{nr}{N_z} \right) \right]$$

$$\tilde{T}[\ell, m, n] = \Delta V \sum_{p, q, r} T[p, q, r] \exp \left[-2\pi i \left(\frac{\ell p}{N_x} + \frac{mq}{N_y} + \frac{nr}{N_z} \right) \right]^1$$

So while generating GRF in Fourier space, we take each $\tilde{T}[\ell, m, n]$ as :

$$\tilde{T}[\ell, m, n] = \sqrt{\frac{V P(k_{\mathbf{n}})}{2}} [a(\ell, m, n) + ib(\ell, m, n)] \quad (28)$$

Where we take $a(\ell, m, n), b(\ell, m, n)$ from a normal distribution $\sim \mathcal{N}(0, 1)$ with zero mean and unit variance. But as $T[p, q, r]$ is a real field we get the symmetry property:

$$\tilde{T}[N_x - \ell, N_y - m, N_z - n] = [\tilde{T}[\ell, m, n]]^* \quad (29)$$

Where ℓ, m, n runs from $0, N_x - 1, 0, N_y - 1$ and $N_z - 1$ respectively. But this gives a condition :

$$\tilde{T}[N_x, N_y, N_z] = \tilde{T}[0, 0, 0] = [\tilde{T}[0, 0, 0]]^* \quad ^2$$

So this states $\tilde{T}[0, 0, 0]$ should be real.

Now 8 cases can arise. We can do the same thing just, if no of points along an axis is even then Nyquist has reality condition. Also going to a plane if the no of points along both axes is even there is also a reality condition at the Nyquist frequency. Also if no of points along all axes are even then :

$$\tilde{T}[N_x/2, N_y/2, N_z/2] = [\tilde{T}[N_x/2, N_y/2, N_z/2]]^* \quad (30)$$

¹ Where we have written

$$\vec{\mathbf{k}}_{\mathbf{n}} \cdot \vec{\mathbf{x}}_{\mathbf{m}} = 2\pi i \left(\frac{\ell p}{N_x} + \frac{mq}{N_y} + \frac{nr}{N_z} \right)$$

² Putting $\ell, m, n = 0, 0, 0$ in Eq. (29) and using periodicity.

making it also real.

If we do have the reality condition then we generate $\tilde{T}[\ell, m, n]$ by:

$$\tilde{T}[\ell, m, n] = \sqrt{V P(k_n)} [a(\ell, m, n)]$$

Where we take $a(\ell, m, n)$ from a normal distribution $\sim \mathcal{N}(0, 1)$ with zero mean and unit variance.

5.1 Units

$\tilde{T}(\vec{k}_n)$ has the unit of $K \text{ Mpc}^3$. Now power spectrum has the unit of $K^2 \text{ Mpc}^3$. So the amplitude A_s has the unit of $K^2 \text{ Mpc}^3$.

6 Smoothing of Fields: [2]

We can replace the field $T(\vec{x})$ by a smoothed field : ¹

$$T_R(\vec{x}) = \int d^3\vec{r} W(\vec{r}) T(\vec{x} + \vec{r}) \quad (31)$$

has been smoothed to scales of length R . Where, $W(\vec{r})$ is a window function.²
Now putting by Eq. (1) we get :

$$\begin{aligned} T_R(\vec{x}) &= \int d^3\vec{r} W(\vec{r}) \int \frac{d^3\vec{k}}{(2\pi)^3} \tilde{T}(\vec{k}) \exp[i\vec{k} \cdot (\vec{x} + \vec{r})] \\ &= \int \frac{d^3\vec{k}}{(2\pi)^3} \tilde{T}(\vec{k}) \exp[i\vec{k} \cdot \vec{x}] \int d^3\vec{r} W(\vec{r}) \exp[i\vec{k} \cdot \vec{r}] \\ &= \int \frac{d^3\vec{k}}{(2\pi)^3} \tilde{T}(\vec{k}) \tilde{W}(-\vec{k}) \exp[i\vec{k} \cdot \vec{x}] \quad ^3 \\ &= \int \frac{d^3\vec{k}}{(2\pi)^3} \tilde{T}_R(\vec{k}) \exp[i\vec{k} \cdot \vec{x}] \end{aligned}$$

Where, we define $\tilde{T}_R(\vec{k}) = \tilde{T}(\vec{k}) \tilde{W}(-\vec{k})$ smoothed field in the Fourier space. So we can see, when we do smoothing, it is basically multiplication in Fourier space. That makes our life very easy. This is not anything new as we can see Eq. (31) is like a convolution in real space, which by convolution theorem is just product in Fourier space.

¹ This is like performing weighted average.

² We can rewrite Eq. (31) in terms of un-normalized window function $\mathcal{W}(\vec{r})$.

$$T_R(\vec{x}) = \frac{\int d^3\vec{r} \mathcal{W}(\vec{r}) T(\vec{x} + \vec{r})}{\int d^3\vec{r} \mathcal{W}(\vec{r})}$$

Where $W(\vec{r}) = \frac{\mathcal{W}(\vec{r})}{\int d^3\vec{r} \mathcal{W}(\vec{r})}$ is the normalized window function.

³ Where

$$\tilde{W}(-\vec{k}) = \int d^3\vec{r} W(\vec{r}) \exp[i\vec{k} \cdot \vec{r}]$$

Fourier transform of $W(\vec{r})$.

Now most commonly used window function is spherical top hat of radius R (normalized by the volume) which is defined as :

$$W(\vec{r}) = A_N \Theta(\vec{R} - \vec{r})^1 = \begin{cases} \frac{3}{4\pi R^3} & \text{for } r \leq R \\ 0 & \text{for } r > R \end{cases} \quad (32)$$

By this we smooth the field at $\vec{x}$ over the sphere of radius R .
Fourier transform of this top hat window function is given by :

$$\tilde{W}(-\vec{k}) = \frac{3}{(kR)^3} \left[\sin(kR) - (kR) \cos(kR) \right]^2 \quad (33)$$

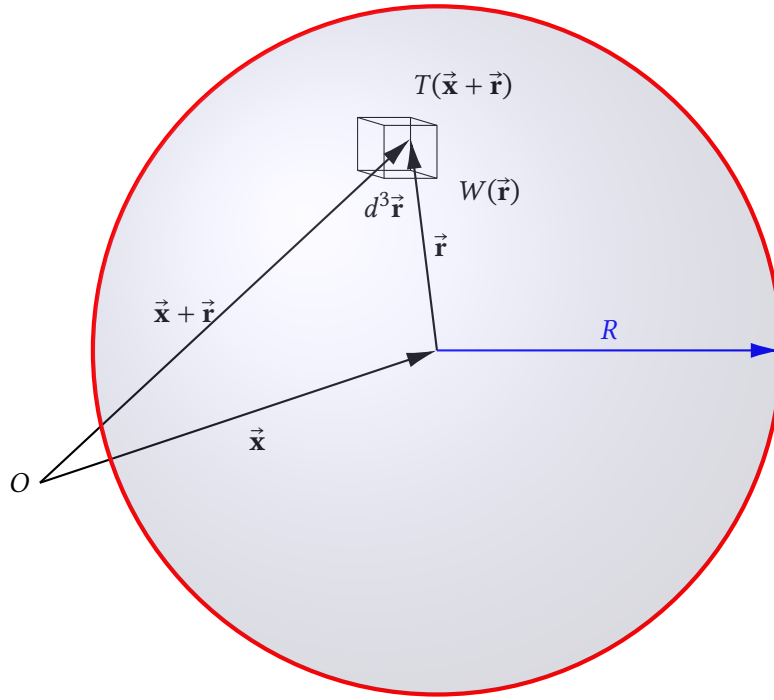


Figure 1: Showing top hat smoothing over length scales R .

¹ $\Theta(\vec{R} - \vec{r})$ is the Heaviside step function which has the definition :

$$\Theta(\vec{R} - \vec{r}) = \begin{cases} 1 & \text{for } |\vec{r}| \leq |\vec{R}| \\ 0 & \text{for } |\vec{r}| > |\vec{R}| \end{cases}$$

$A_N(\vec{r})$ is the normalization value which is given by :

$$A_N = \left[\int d^3\vec{r} \Theta(\vec{R} - \vec{r}) \right]^{-1} = \left[\int_0^R \int_0^\pi \int_0^{2\pi} r^2 dr \sin \theta d\theta d\phi \right]^{-1} = \left[\frac{4}{3} \pi R^3 \right]^{-1} = \frac{3}{4\pi R^3}$$

to make $W(\vec{r})$ normalized.

² See Appendix appendix B.2

We can also use Gaussian smoothing function which is given by : [5]

$$W(\vec{r}) = \frac{1}{(2\pi)^{3/2} R^3} \exp\left[-\frac{|\vec{r}|^2}{2R^2}\right] \quad {}^1 \quad (34)$$

Fourier transform of this is given by $\tilde{W}(-\vec{k}) = \exp\left[-\frac{k^2 R^2}{2}\right] \quad {}^2$

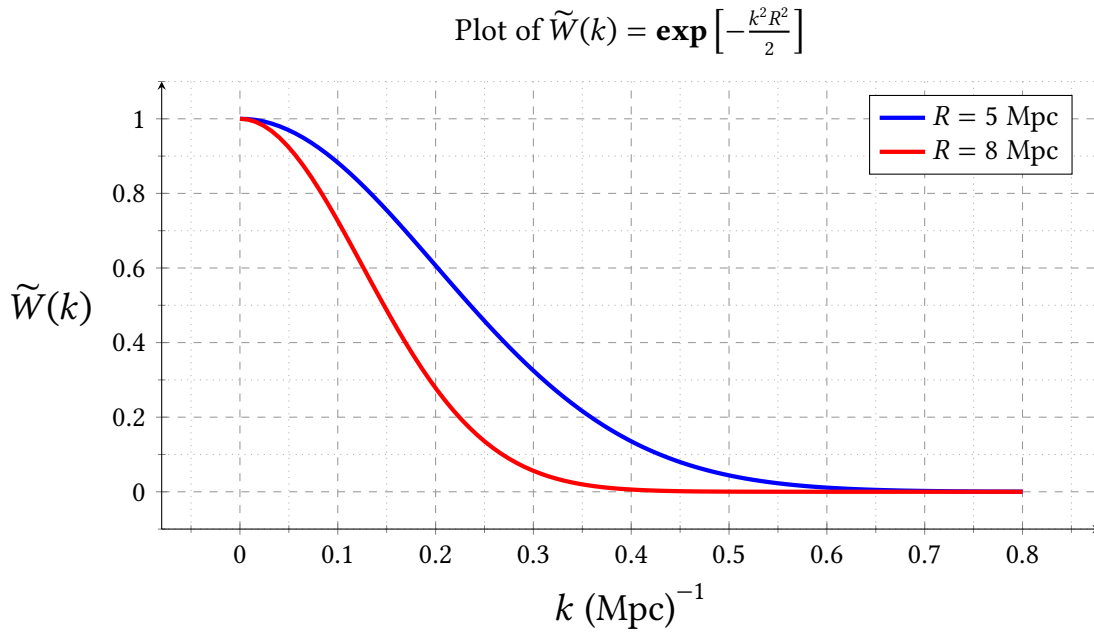


Figure 2: Plot of Window function $\tilde{W}(k) = \exp\left[-\frac{k^2 R^2}{2}\right]$ for different values of R .

¹ Here the normalization is given by :

$$\begin{aligned} A_N &= \left\{ \int d^3\vec{r} \exp\left[-|\vec{r}|^2/(2R^2)\right] \right\}^{-1} \\ &= \left\{ \iiint dx dy dz \exp\left[-(x^2 + y^2 + z^2)/(2R^2)\right] \right\}^{-1} \\ &= \left\{ \int dx \exp\left[-x^2/(2R^2)\right] \right\}^{-3} \\ &= \left\{ \sqrt{2\pi R^2} \right\}^{-3} \\ &= \frac{1}{(2\pi)^{3/2} R^3} \end{aligned}$$

Where we have used the formulae for standard integral :

$$\int_{-\infty}^{\infty} dx \exp[-\alpha x^2] = \sqrt{\frac{\pi}{\alpha}}$$

² See Appendix appendix B.

6.1 Smoothing of fields in 2D

Following the same things over the last pages, we can define a top hat filter in 2D by :

$$W(\vec{\mathbf{r}}) = \begin{cases} \frac{1}{\pi R^2} & \text{for } r \leq R \\ 0 & \text{for } r > R \end{cases} \quad (35)$$

The Fourier transform of this is given by :

$$\begin{aligned} \tilde{W}(-\vec{\mathbf{k}}) &= \int d^2\vec{\mathbf{r}} W(\vec{\mathbf{r}}) \exp[i\vec{\mathbf{k}} \cdot \vec{\mathbf{r}}] \\ &= \iint r dr d\theta W(r) \exp[-ikr \cos \theta] \\ &= \int_0^\infty r dr W(r) \int_0^{2\pi} d\theta \exp[-ikr \cos \theta] \\ &= \int_0^\infty r dr W(r) \{2\pi J_0(kr)\}^1 \\ &= \frac{2}{R^2} \int_0^R r dr J_0(kr) \\ &= \frac{2 J_1(kR)}{kR}^2 \end{aligned}$$

By the same token, we define 2D Gaussian smoothing function as :

$$W(\vec{\mathbf{r}}) = \frac{1}{2\pi R^2} \exp\left[-\frac{|\vec{\mathbf{r}}|^2}{2R^2}\right] \quad (36)$$

Fourier transform of this is given by $\tilde{W}(-\vec{\mathbf{k}}) = \exp\left[-\frac{k^2 R^2}{2}\right]$

¹ We have used the standard formula :

$$\int_0^{2\pi} e^{-iz \cos \theta} d\theta = 2\pi J_0(z)$$

where $J_0(z)$ is the Bessel function of the first kind of order zero.^[1]

² See Appendix appendix B.3 for more details.

A Discrete Fourier Transform & Statistical Homogeneity

We take a function $f(x)$ to ranged over 0 to L , which is taken to be periodic outside the above stated range. We sample the data at N discrete points along the x axis given by $x_n = n\Delta x$, $\Delta x = L/N$, where $n = 0, 1, \dots, N-1$. We denote $f(x)$ at $x = x_n$ as $f[n]$. Then consider the Discrete Fourier transformations (in 1D) given by :

$$f[n] = \frac{1}{L} \sum_{m=0}^{N-1} \tilde{f}[m] \exp\left[\frac{2\pi i m n}{N}\right] \quad (37)$$

$$\tilde{f}[m] = \frac{L}{N} \sum_{n=0}^{N-1} f[n] \exp\left[-\frac{2\pi i m n}{N}\right] \quad (38)$$

Now Discrete Fourier transformation assumes that our function $f[n]$ is periodic outside the range, i.e. $f[n+N] = f[n]$.¹ For example for $N = 3$ we have values $f[0], f[1], f[2]$. Now by this property we have :

$$\begin{aligned} f[3] &= f[0] \\ f[4] &= f[1] \\ f[5] &= f[2] \end{aligned}$$

Also for negative indices

$$\begin{aligned} f[-3] &= f[0] \\ f[-2] &= f[1] \\ f[-1] &= f[2] \end{aligned}$$

Those negative indices corresponds to spatial point $x_{-n} = (-n)\Delta x$.

Now this makes a very interesting fact, that in Fourier space :

$$\begin{aligned} \tilde{f}[m + \alpha N] &= \left(\frac{L}{N}\right) \sum_{n=0}^{N-1} f[n] \exp\left[-\frac{2\pi i (m + \alpha N)n}{N}\right] \\ &= \left(\frac{L}{N}\right) \sum_{n=0}^{N-1} f[n] \exp\left[-\frac{2\pi i m n}{N}\right] \times \exp[-2\pi i \alpha n] \\ &= \left(\frac{L}{N}\right) \sum_{n=0}^{N-1} f[n] \exp\left[-\frac{2\pi i m n}{N}\right] \stackrel{2}{=} \tilde{f}[m] \end{aligned}$$

So in Fourier space the function $\tilde{f}$ also becomes periodic.

Let us evaluate 2 point correlation function in Fourier space, i.e. $\langle \tilde{f}(m) [\tilde{f}(m')]^* \rangle$.

$$\langle \tilde{f}(m) [\tilde{f}(m')]^* \rangle = \left(\frac{L}{N}\right)^2 \sum_{n=0}^{N-1} \sum_{n'=0}^{N-1} \langle f[n] f[n'] \rangle \exp\left[\frac{2\pi i (m'n' - mn)}{N}\right]$$

Now if we impose statistical homogeneity then $\langle f[n] f[n'] \rangle = \xi[n - n']$ ³

Then the last equation becomes :

$$\langle \tilde{f}(m) [\tilde{f}(m')]^* \rangle = \left(\frac{L}{N}\right)^2 \sum_{n=0}^{N-1} \sum_{n'=0}^{N-1} \xi[n - n'] \exp\left[\frac{2\pi i (m'n' - mn)}{N}\right]$$

¹ More precisely, for any index p we have $f[p \bmod N] = f[p]$, or alternatively $f[n + \alpha N] = f[n] \forall \alpha \in \mathbb{Z}$.

² As α and n both are integers, then $\exp[-2\pi i \alpha n] = 1 \forall \alpha \in \mathbb{Z}$.

³ ξ is the 2 point correlation function in real space. As f is periodic this forces ξ also to be periodic over the indices. So by the same token we have $\xi[n + \alpha N] = \xi[n] \forall \alpha \in \mathbb{Z}$.

Now changing the order of summation and making the substitution $n - n' = n''$ gives :

$$\begin{aligned}
 \left\langle \tilde{f}(m) [\tilde{f}(m')]^* \right\rangle &= \left(\frac{L}{N} \right)^2 \sum_{n'=0}^{N-1} \exp \left[\frac{2\pi i m' n'}{N} \right] \sum_{n''=-n'}^{N-1-n'} \xi[n''] \exp \left[-\frac{2\pi i m(n' + n'')}{N} \right] \\
 &= \left(\frac{L}{N} \right)^2 \sum_{n'=0}^{N-1} \exp \left[\frac{2\pi i (m' - m)n'}{N} \right] \sum_{n''=-n'}^{N-1-n'} \xi[n''] \exp \left[-\frac{2\pi i m n''}{N} \right] \\
 &= \left(\frac{L}{N} \right)^2 \sum_{n'=0}^{N-1} \exp \left[\frac{2\pi i (m' - m)n'}{N} \right] \sum_{n''=0}^{N-1} \xi[n''] \exp \left[-\frac{2\pi i m n''}{N} \right] \text{ }^1 \\
 &= \left(\frac{L}{N} \right)^2 \cdot \left\{ N \delta_{mm'} \right\} \sum_{n''=0}^{N-1} \xi[n''] \exp \left[-\frac{2\pi i m n''}{N} \right] \\
 &= L \delta_{mm'} \left\{ \left(\frac{L}{N} \right) \sum_{n''=0}^{N-1} \xi[n''] \exp \left[-\frac{2\pi i m n''}{N} \right] \right\} \\
 &= L \delta_{mm'} P(m)
 \end{aligned}$$

$$\left\langle \tilde{f}(m) [\tilde{f}(m')]^* \right\rangle = L \delta_{mm'} P(m) \tag{39}$$

$$P(m) = \frac{1}{L} \left\langle |\tilde{f}(m)|^2 \right\rangle \tag{40}$$

$$P(m) = \left(\frac{L}{N} \right) \sum_{n''=0}^{N-1} \xi[n''] \exp \left[-\frac{2\pi i m n''}{N} \right] \tag{41}$$

¹ If you are paying attention, we have changed the range of n'' from $-n'$ to $N - 1 - n''$ to 0 to $N - 1$. Why this is right, has been discussed in appendix A.1.

A.1 Some Proofs

- Symmetry of $\tilde{f}$

$$\begin{aligned}
 \tilde{f}[N-m] &= \left(\frac{L}{N}\right) \sum_{n=0}^{N-1} f[n] \exp\left[-\frac{2\pi i(N-m)n}{N}\right] \\
 &= \left(\frac{L}{N}\right) \sum_{n=0}^{N-1} f[n] \exp\left[\frac{2\pi i m n}{N}\right] \times \exp[-2\pi i n] \\
 &= \left(\frac{L}{N}\right) \sum_{n=0}^{N-1} f[n] \exp\left[\frac{2\pi i m n}{N}\right]
 \end{aligned}$$

If $f[n]$ is real then $\tilde{f}[N-m] = [\tilde{f}[m]]^*$ ($m = 0, 1, \dots, N-1$).

We could extend this idea in higher dimensions.

$$\begin{aligned}
 \tilde{f}[\mathbf{N} - \mathbf{n}] &= \sum_{\mathbf{m}} f(\mathbf{m}) \exp\left[-2\pi i \sum_k \frac{(N_k - n_k)m_k}{N_k}\right] \\
 &= \sum_{\mathbf{m}} f(\mathbf{m}) \exp\left[2\pi i \sum_k \frac{n_k m_k}{N_k}\right] \times \exp\left[-2\pi i \sum_k m_k\right] \\
 &= \sum_{\mathbf{m}} f(\mathbf{m}) \exp\left[2\pi i \sum_k \frac{n_k m_k}{N_k}\right] \quad \textcolor{red}{1}
 \end{aligned}$$

• **Proof:** $\sum_{n=0}^{N-1} \exp\left[\frac{2\pi i(m' - m)n}{N}\right] = N \delta_{mm'}$

For $m = m'$ the exponential factor is 1 for all n s and the sum $\sum_0^{N-1} 1 = N$.
For $m \neq m'$ case we have :

$$\begin{aligned}
 &\sum_{n=0}^{N-1} \exp\left[\frac{2\pi i(m' - m)n}{N}\right] \\
 &= \sum_{n=0}^{N-1} \omega^{yn} \quad \textcolor{red}{2} \\
 &= 1 + \omega^y + \omega^{2y} + \dots + \omega^{(N-1)y} \\
 &= \frac{1 - \omega^N}{1 - \omega} = 0
 \end{aligned}$$

So combining we have :

$$\sum_{n=0}^{N-1} \exp\left[\frac{2\pi i(m' - m)n}{N}\right] = N \delta_{mm'} = \begin{cases} N & \text{for } m = m' \\ 0 & \text{for } m \neq m' \end{cases}$$

We can extend this idea for higher dimensions.

$$\sum_{\mathbf{n}} \exp\left[2\pi i \sum_k \frac{(m'_k - m_k)n_k}{N_k}\right]$$

¹ Where $\mathbf{N} = N_1, N_2, \dots$ same goes for $\mathbf{n}, \mathbf{m}$. And $\sum_{\mathbf{m}} = \prod_k \sum_{m_k} = \sum_{m_1} \sum_{m_2} \dots$

² We have taken $\omega = \exp\left[\frac{2\pi i}{N}\right]$ and $y = m' - m$

$$\begin{aligned}
&= \prod_k \left\{ \sum_{n_k=0}^{N_k-1} \exp \left[\frac{2\pi i(m'_k - m_k)n_k}{N_k} \right] \right\} \\
&= \prod_k \{N_k \delta_{m_k m'_k}\} \\
&= \left(\prod_k N_k \right) \left(\prod_k \delta_{m_k m'_k} \right) \\
&= \left(\prod_k N_k \right) \delta_{\mathbf{m} \mathbf{m}'}
\end{aligned}$$

• **Proof :** $\sum_{n=0}^{N-1} \xi[n] \exp \left[-\frac{2\pi i m n}{N} \right] = \sum_{n=-n'}^{N-1-n'} \xi[n] \exp \left[-\frac{2\pi i m n}{N} \right]$

So our claim is shifting the indices by n' points does not change the sum. At a first glance, we can say this is happening due to the fact that the function is periodic.

We write $\omega = \exp \left[\frac{2\pi i}{N} \right]$ and expand out the terms. Here while doing the summation n' and m both being fixed and the can take any integer values ranges form 0 to $N - 1$.

$$\begin{aligned}
&\sum_{n=-n'}^{N-1-n'} \xi[n] \exp \left[-\frac{2\pi i m n}{N} \right] \\
&= \sum_{n=-n'}^{N-1-n'} \xi[n] \omega^{-m n} \\
&= \xi[-n'] \omega^{m n'} + \xi[-n' + 1] \omega^{m(1-n')} + \dots + \xi[-1] \omega^m + \\
&\quad + \xi[0] + \xi[1] \omega^{-m} + \dots + \xi[N - 1 - n'] \omega^{m(1+n'-N)} \\
&= \omega^{-N m} [\xi[-n'] \omega^{m n'} + \xi[-n' + 1] \omega^{m(1-n')} + \dots + \xi[-1] \omega^m] + \\
&\quad + \xi[0] + \xi[1] \omega^{-m} + \dots + \xi[N - 1 - n'] \omega^{m(1+n'-N)} \\
&= [\xi[-n'] \omega^{-m(N-n')} + \xi[-n' + 1] \omega^{-m(N-n'+1)} + \dots + \xi[-1] \omega^{-m(N-1)}] + \\
&\quad + \xi[0] + \xi[1] \omega^{-m} + \dots + \xi[N - 1 - n'] \omega^{m(1+n'-N)} \\
&= [\xi[N - n'] \omega^{-m(N-n')} + \xi[N - n' + 1] \omega^{-m(N-n'+1)} + \dots + \\
&\quad + \xi[N - 1] \omega^{-m(N-1)}] + \xi[0] + \xi[1] \omega^{-m} + \dots + \xi[N - 1 - n'] \omega^{m(1+n'-N)} \\
&= \xi[0] + \xi[1] \omega^{-m} + \dots + \xi[N - 1 - n'] \omega^{m(1+n'-N)} + [\xi[N - n'] \omega^{-m(N-n')} + \\
&\quad + \xi[N - n' + 1] \omega^{-m(N-n'+1)} + \dots + \xi[N - 1] \omega^{-m(N-1)}] \\
&= \sum_{n=0}^{N-1} \xi[n] \omega^{-m n} \\
&= \sum_{n=0}^{N-1} \xi[n] \exp \left[-\frac{2\pi i m n}{N} \right]
\end{aligned}$$

Where we have used the periodicity in $\xi[n]$ for the terms in which $n < 0$ and also multiplied with $\omega^{-N m}(= 1)$. As you can see for $n' = 0$ its the same . But for $n' \neq 0$ what we do is, expand the terms and for the terms where in the ξ function the argument $\xi[n]$, the n becomes negative we shift them by N index, and multiply them by $\omega^{-N m}$, and finally we take them after the term $\xi[N - 1 - n']$ and we got the result.

Now we do in higher dimension (say d dimensional space) :

$$\sum_{\mathbf{n}=-\mathbf{n}'}^{N-\mathbf{n}'-1} \xi[\mathbf{n}] \exp \left[-2\pi i \sum_k \frac{m_k n_k}{N_k} \right]$$

$$= \left\{ \prod_{k=1}^d \sum_{n_k=-n'_k}^{N_k-n'_k-1} \right\} \xi[\mathbf{n}] \exp \left[-2\pi i \sum_k \frac{m_k n_k}{N_k} \right]$$

Now we start the sum from the last dimension $k = d$, and replace the sum

$\sum_{n_d=-n'_d}^{N_d-n'_d-1}$ by $\sum_{n_d=0}^{N_d-1}$ as we see it in the previous case and we use :

$$\xi[n_1, n_2, \dots, n_d] = \xi[n_1, n_2, \dots, N_d + n_d]$$

In fact we can do this along any dimension(s). Then do the 2nd last dimension and so on. And in the end we got the result.

To be precise we can write $\xi[\mathbf{n}] = \xi[\mathbf{n} + \boldsymbol{\alpha} \circ \mathbf{N}]$ ¹

$$\begin{aligned} & \left\{ \prod_{k=1}^d \sum_{n_k=-n'_k}^{N_k-n'_k-1} \right\} \xi[\mathbf{n}] \exp \left[-2\pi i \sum_k \frac{m_k n_k}{N_k} \right] \\ &= \sum_{n_1=-n'_1}^{N_1-n'_1-1} \cdots \sum_{n_d=-n'_d}^{N_d-n'_d-1} \xi[\mathbf{n}] \exp \left[-2\pi i \sum_k \frac{m_k n_k}{N_k} \right] \\ &= \sum_{n_1=-n'_1}^{N_1-n'_1-1} \cdots \left[\sum_{n_d=-n'_d}^{N_d-n'_d-1} \xi[\mathbf{n}] \exp \left[-2\pi i \frac{m_d n_d}{N_d} \right] \right] \exp \left[-2\pi i \sum_{k \neq d} \frac{m_k n_k}{N_k} \right] \\ &= \sum_{n_1=-n'_1}^{N_1-n'_1-1} \cdots \left[\sum_{n_d=0}^{N_d-1} \xi[\mathbf{n}] \exp \left[-2\pi i \frac{m_d n_d}{N_d} \right] \right] \exp \left[-2\pi i \sum_{k \neq d} \frac{m_k n_k}{N_k} \right] \\ &= \sum_{n_d=-n'_d}^{N_d-1} \sum_{n_1=-n'_1}^{N_1-n'_1-1} \cdots \left[\sum_{n_{d-1}=-n'_{d-1}}^{N_{d-1}-n'_{d-1}-1} \xi[\mathbf{n}] \exp \left[-2\pi i \frac{m_{d-1} n_{d-1}}{N_{d-1}} \right] \right] \exp \left[-2\pi i \sum_{k \neq d-1} \frac{m_k n_k}{N_k} \right] \\ &= \sum_{n_d=-n'_d}^{N_d-1} \sum_{n_1=-n'_1}^{N_1-n'_1-1} \cdots \left[\sum_{n_{d-1}=0}^{N_{d-1}-1} \xi[\mathbf{n}] \exp \left[-2\pi i \frac{m_{d-1} n_{d-1}}{N_{d-1}} \right] \right] \exp \left[-2\pi i \sum_{k \neq d-1} \frac{m_k n_k}{N_k} \right] \\ &\quad \vdots \\ &\quad \vdots \\ &\quad \vdots \\ &= \sum_{n_1=0}^{N_1-1} \cdots \sum_{n_d=0}^{N_d-1} \xi[\mathbf{n}] \exp \left[-2\pi i \sum_k \frac{m_k n_k}{N_k} \right] \\ &= \left\{ \prod_{k=1}^d \sum_{n_k=0}^{N_k-1} \right\} \xi[\mathbf{n}] \exp \left[-2\pi i \sum_k \frac{m_k n_k}{N_k} \right] \end{aligned}$$

¹ Where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \dots) \forall \alpha_k \in \mathbb{Z}$, $\boldsymbol{\alpha} \circ \mathbf{N} = (\alpha_1 N_1, \alpha_2 N_2, \dots)$ with $(\boldsymbol{\alpha} \circ \mathbf{N})_k = \alpha_k N_k$

B Additional Evaluations:

B.1 n - dimensional Gaussian and it's Fourier Transformation :

B.1.1 1D Gaussian :

$$W(x) = \frac{1}{\sqrt{2\pi R^2}} \exp\left[-\frac{x^2}{2R^2}\right]$$

where $1/\sqrt{2\pi R^2}$ factor does the normalization such that $\int_{-\infty}^{\infty} W(x) dx = 1$.¹

B.1.2 Fourier Transform :

Now taking Fourier transform we get :

$$\begin{aligned}\tilde{W}(-k) &= \int_{-\infty}^{\infty} dx W(x) \exp[ikx] \\ &= \frac{1}{\sqrt{2\pi R^2}} \int_{-\infty}^{\infty} dx \exp\left[-\frac{x^2}{2R^2} + ikx\right] \quad ^2 \\ &= \frac{1}{\sqrt{2\pi R^2}} \cdot \sqrt{2\pi R^2} \cdot \exp\left[-\frac{k^2}{4 \cdot \frac{1}{2R^2}}\right] \\ &= \exp\left[-\frac{k^2 R^2}{2}\right]\end{aligned}$$

So we see Fourier transform of a Gaussian is also a Gaussian.

$$\boxed{\tilde{W}(k) = \exp\left[-\frac{k^2 R^2}{2}\right]} \quad (42)$$

B.1.3 n - dimensional Gaussian :

In a very similar way, we define n dimensional Gaussian as :

$$W(\vec{x}) = \frac{1}{(2\pi R^2)^{n/2}} \exp\left[-\frac{1}{2R^2} \sum_{j=1}^n x_j^2\right] \quad ^3$$

¹ This is because as :

$$\int_{-\infty}^{\infty} dx \exp\left[-\frac{x^2}{2R^2}\right] = \sqrt{2\pi R^2}$$

² Where we have used the formulae for standard integral :

$$\int_{-\infty}^{\infty} dx \exp[-\alpha x^2 \pm i\beta x] = \sqrt{\frac{\pi}{\alpha}} \exp\left[-\frac{\beta^2}{4\alpha}\right]$$

³ Where $\vec{x} = (x_1, x_2, \dots, x_n)$.

Now Fourier transform gives :

$$\begin{aligned}
 \tilde{W}(-\vec{k}) &= \int d^n \vec{x} W(\vec{x}) \exp[i \vec{k} \cdot \vec{x}] \\
 &= \int \prod_{j=1}^n dx_j W(\vec{x}) \exp\left[i \sum_{j=1}^n k_j x_j\right] \\
 &= \frac{1}{(2\pi R^2)^{n/2}} \prod_{j=1}^n \left\{ \int dx_j \exp\left[-\frac{x_j^2}{2R^2} + ik_j x_j\right] \right\} \\
 &= \frac{1}{(2\pi R^2)^{n/2}} \prod_{j=1}^n \left\{ \sqrt{2\pi R^2} \exp\left[-\frac{k_j^2 R^2}{2}\right] \right\} \\
 &= \exp\left[-\left(\sum_{j=1}^n k_j^2\right) \frac{1}{2R^2}\right] \\
 &= \exp\left[-\frac{|\vec{k}|^2}{2R^2}\right]
 \end{aligned}$$

B.2 Fourier Transform of Top Hat window function :

$$\begin{aligned}
 \tilde{W}(-\vec{k}) &= \int d^3 \vec{r} W(\vec{r}) \exp[i \vec{k} \cdot \vec{r}] \\
 &= \iiint r^2 dr \sin \theta d\theta d\phi W(r) \exp[-ikr \cos \theta] \\
 &= 4\pi \int_0^\infty r^2 dr W(r) \frac{\sin kr}{kr} \\
 &= \frac{3}{(kR)^3} \int_0^{kR} u du \sin u \\
 &= \frac{3}{(kR)^3} [-u \cos u + \sin u] \Big|_0^{kR} \\
 &= \frac{3}{(kR)^3} [\sin(kR) - (kR) \cos(kR)]
 \end{aligned}$$

We write $y = kR$. So the function becomes :

$$\tilde{W}(k) = \frac{3}{y^3} [\sin y - y \cos y] \quad (43)$$

B.3 Bessel Functions Recursion :

The derivatives of Bessel functions are given by :

$$\boxed{\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)} \quad (44)$$

As in Sec. 6.1 -

$$\begin{aligned}
 &\int_0^R r dr J_0(kr) \\
 &= \frac{1}{k^2} \int_0^{kR} u du J_0(u) \text{ (substitution } u = kr)
 \end{aligned}$$

$$\text{Plot of } \tilde{W}(k) = \frac{3}{y^3} [\sin y - y \cos y], \text{ with } y = kR$$

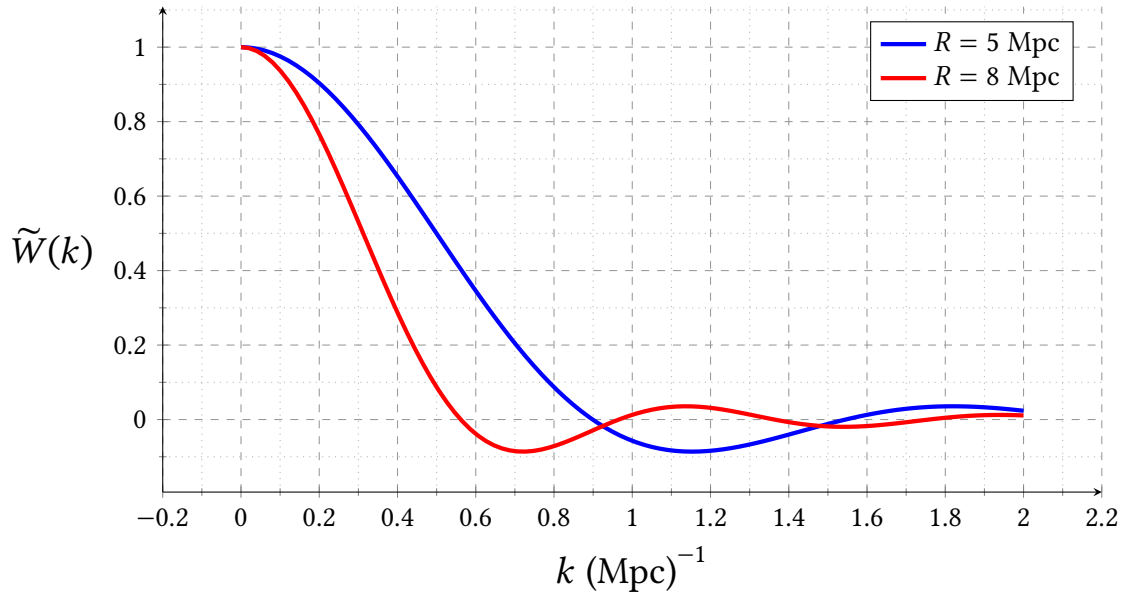


Figure 3: Plot of Window function $\tilde{W}(k) = \frac{3}{y^3} [\sin y - y \cos y]$ for different values of R .

$$\begin{aligned} &= \frac{1}{k^2} \int_0^{kR} du [u J_0(u)] \\ &= \frac{1}{k^2} \int_0^{kR} d[u J_1(u)] \\ &= \frac{1}{k^2} [u J_1(u)] \Big|_0^{kR} \\ &= \frac{R^2 J_1(kR)}{kR} \end{aligned}$$

Here J_1 is the Bessel function of the first kind of order one.

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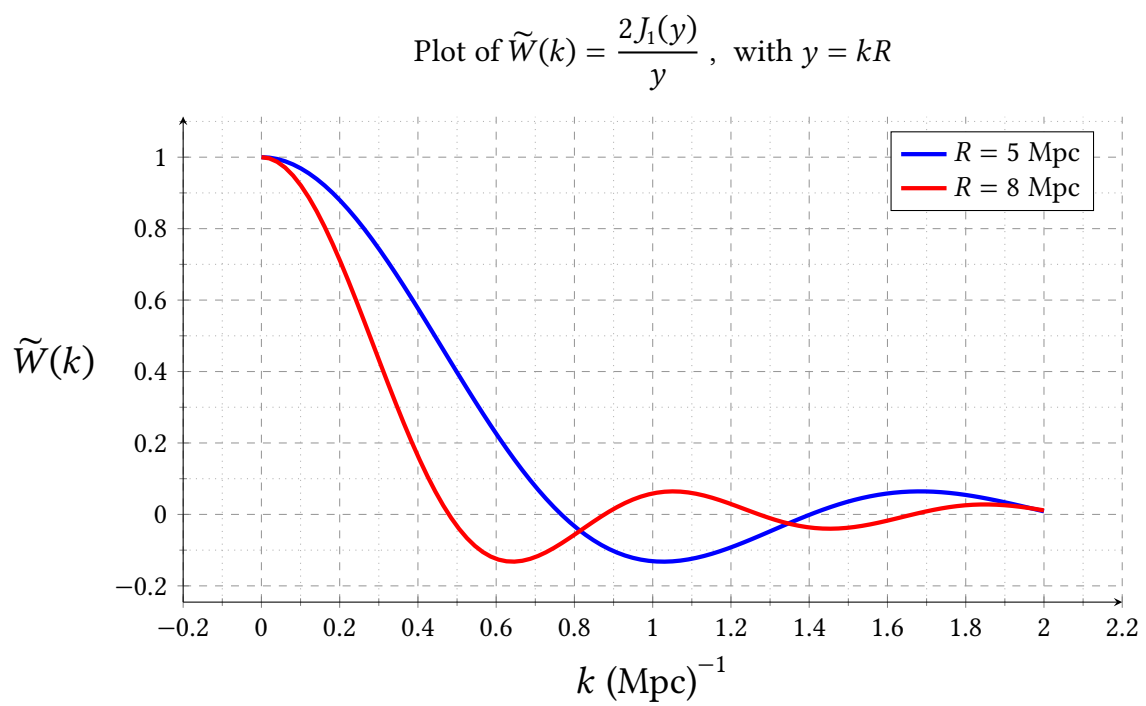


Figure 4: Plot of Window function $\tilde{W}(k) = \frac{2J_1(y)}{y}$ for different values of R .